# CYBERSTER RED TEAMING

## INTERSHIP TASK REPORT

*Network Enumeration & Service Vulnerability Discovery*

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|---|---|
| Name | Muhammad Hozaifa Naeem |
| Roll Number | CSI-B1-049 |
| Internship Program | Cyberster Cybersecurity Internship |
| Target Domain | Hackthissite.org , php.vulntest.org |
| Task Title | Network Enumeration & Service Vulnerability Discovery |
| Classification | Confidential-For Targeted network scanning |

# Network Enumeration & Service Vulnerability Discovery

## Task 01: Advanced Network Scanning with Nmap

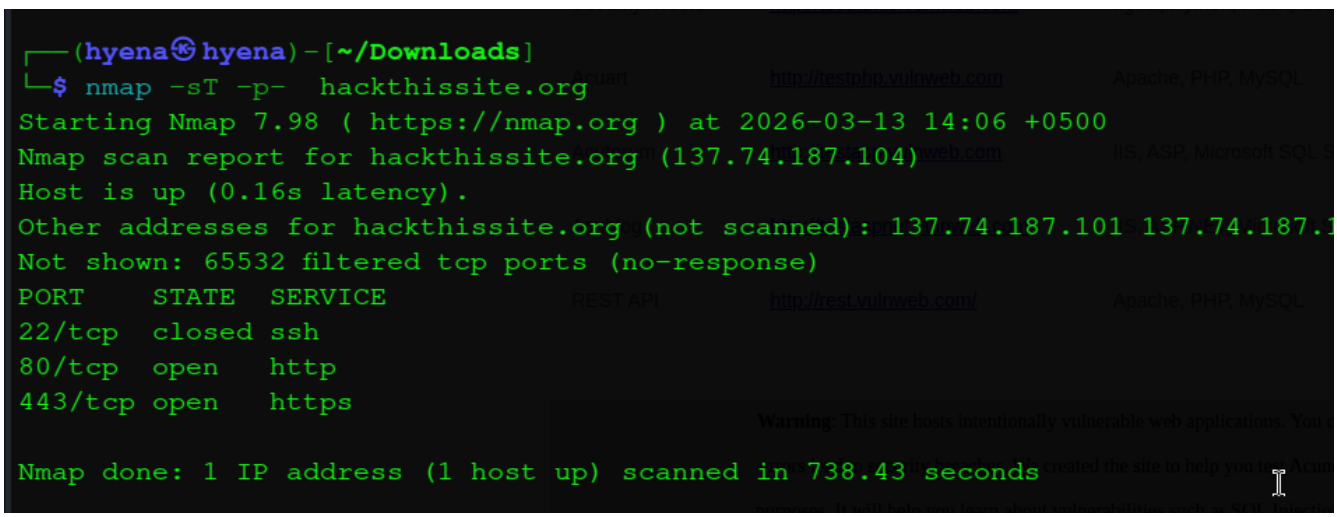
I Perform TCP Connect scans (-sT) and SYN Stealth scans (-sS) and compare the results in terms of speed and visibility.

## 1.TCP Connect Scan (-sT)

### *Command:*

nmap -sT -p- -T3 hackthissite.org

### *POC:*



```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sT -p- hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:06 +0500
Nmap scan report for hackthissite.org (137.74.187.104)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.1
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE  SERVICE
22/tcp   closed ssh
80/tcp   open   http
443/tcp  open   https

Nmap done: 1 IP address (1 host up) scanned in 738.43 seconds
```

### *Explanation:*

➔ **-sT (TCP Connect Scan)**

This flag tells Nmap to perform a TCP Connect scan, where it completes the full TCP three-way handshake to determine whether a port is open.

➔ **-p-**

This flag instructs Nmap to scan all possible TCP ports (1–65535) instead of scanning only the most commonly used ports.

# 2.SYN Stealth Scan (-sS)

## *Command:*

sudo nmap -sS -p- -T3 hackthisite.org

## *Explanation:*

➔ ***-sS (SYN Stealth Scan)***
This flag enables SYN scanning, where Nmap sends only the SYN packet and stops the connection before completing the full TCP handshake, making the scan stealthier.

➔ ***-p-***
This option tells Nmap to scan all 65,535 ports on the target system rather than only the most common ports.

➔ ***-T3***
This timing option sets the scanning speed to a normal level, ensuring a balance between accuracy and performance.

## *POC:*

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sS -p- hackthissite.org
[sudo] password for hyena:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:26 +0500
Nmap scan report for hackthissite.org (137.74.187.100)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.10
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE  SERVICE
22/tcp   closed ssh
80/tcp   open   http
443/tcp  open   https

Nmap done: 1 IP address (1 host up) scanned in 236.98 seconds
```

# 3.UDP Scan on Top 100 Ports

## *Command*

sudo nmap -sU --top-ports 100 -T4 hackthissite.org

## *Explanation*

➔ ***sudo***

This command runs Nmap with administrator privileges, which are required for UDP scans because they involve sending low-level network packets.

➔ ***-sU (UDP Scan)***

This flag tells Nmap to perform a UDP scan, which checks UDP ports by sending UDP packets and analyzing the responses from the target system.

➔ ***--top-ports 100***

This option instructs Nmap to scan the **100 most common UDP ports**, which usually Include important services like DNS, SNMP, and DHCP.

➔ ***-T4/5***

This timing template sets the scan speed to a aggressive and insan scanning at very fast speed.

## *POC*

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sU --top-ports 100 -T5 hackthissite.org

[sudo] password for hyena:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:28 +0500
Nmap scan report for hackthissite.org (137.74.187.102)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.103 137.74.187.104 1
All 100 scanned ports on hackthissite.org (137.74.187.102) are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 8.19 seconds

┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sU --top-ports 100 -T4 hackthissite.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:29 +0500
Nmap scan report for hackthissite.org (137.74.187.100)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.102 137.74.187.103 1
rDNS record for 137.74.187.100: 100.187.74.137.in-addr.arpa
All 100 scanned ports on hackthissite.org (137.74.187.100) are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 5.94 seconds
```

| Timing | Name | Speed | Usage |
|---|---|---|---|
| T0 | Paranoid | Extremely slow | IDS evasion |
| T1 | Sneaky | Very slow | Avoid detection |
| T2 | Polite | Slow | Low network impact |
| T3 | Normal | Default | Balanced |
| T4 | Aggressive | Fast | CTF / lab |
| T5 | Insane | Very fast | Risky |

➔ ***-T0 (Paranoid)***
This timing option performs an extremely slow scan to avoid detection by intrusion detection systems.

➔ ***-T1 (Sneaky)***
This timing template runs a very slow scan designed to reduce the chance of being detected by security monitoring systems.

➔ ***-T2 (Polite)***
This option slows down the scan to reduce the load on the network and the target system.

➔ ***-T3 (Normal)***
This is the default scanning speed that balances reliability and performance.

➔ ***-T4 (Aggressive)***
This timing option speeds up the scan significantly but increases the chances of detection.

➔ ***-T5 (Insane)***
This option performs extremely fast scans, which can cause unreliable results and is easily detectable by security systems.

## ***Conclusion***

UDP scanning is slower than TCP scanning because UDP does not use a handshake and often does not respond to probes. Timing templates in Nmap control the speed of scanning, where slower scans improve stealth and reliability, while faster scans increase detection risk and may produce inaccurate results.